QUANT

Permutation, Combination & Probability

Assignment 01

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P & C Assignments 01

1. How many even numbers of four digits can be formed with the digits 0, 1, 2, 3, 4, 5 and 6; no digit being used more than once?

1. 120 2. 300 3. 420 4. 500

2. How many numbers of four digits greater than 2,300 can be formed with digits 0, 1, 2, 3, 4, 5 & 6; no digit being repeated in any number?

1. 480 2. 80 3. 560 4. None of these

3. How many different words can be formed the letters of the word 'UNIVERSITY' so that all the vowels are together.

1. 120960 2. 60480 3. 12 4. None of these

4. In how many ways can the letters of the word of the word 'DIRECTOR' be arranged so that all the three vowels are never together

1. 360 2. 18000 3. 20160 4. 720

5. How many numbers of 4 digits can be formed with the digits 0, 1, 2 and 3? Find the sum of these numbers

1. 24, 38664 2. 18, 38664 3. 24, 39664 4. 18, 39664

6. Four dice are rolled. The number of possible out comes in which at least one dice shows 2 is

1. 1,296 2. 671 3. 625 4. None of these

7. How many 10 digit numbers can be written by using the digit 1 and 2?

1. ${}^{10}C_1 + {}^{9}C_2$ 2. 2^{10} 3. ${}^{10}C_2$ 4. 10!

8. In two successive throws of a pair of dice, determine the probability of getting a total of 8, each time.

1. $\frac{25}{1296}$ 2. $\frac{20}{1296}$ 3. $\frac{16}{1296}$ 4. $\frac{30}{1296}$

9. In a single throw of three dice, find the probability of getting a total of 17 or 18.

1. $\frac{1}{6}$ 2. $\frac{1}{36}$ 3. $\frac{5}{216}$ 4. $\frac{1}{54}$

10. A polygon has 44 diagonals, the number of its sides is

1. 9 2. 10 3. 11 4. 12

Answer key & Explanations

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1.	3	Here no. of digits = 7 Case – I where 'O' occurs at unit place Thousand Hundred Tens Unit ↓ 0 Unit place can be filled in 1 way (0) ; Ten's place can be filled in 6 ways Hundred's place can be filled in 5 ways ; Thousand's place can be filled in 4 ways ; Using fundamental principle of multiplication the required no. = $1\times6\times5\times4=120$; Case – II When 0 does not occur at unit place Thousand Hundred Tens Unit ↓ 2, 4, 6 Unit place can be filled in 3 ways (2, 4, 6) ; Thousand place can be filled in 5 ways; (one of the six digits other the zero) ; Hundred place can be filled in 5 ways; Ten's place can be filled in 4 ways Required no. ways = 3×5×5×4=300. Total no. of numbers = 120+300=420
2.	3	Case – I When 2 occurs at thousand place Thousand place can be filled up by 2 in 1 way Hundred place can be filled up by any of the four digits i.e. 3, 4, 5 and 6 in 4 ways; Tens place can be filled in 5 ways ; unit place can be filled in 4 ways; using fundamental principle of multiplication, the required no. 1×4×5×4=80 Case – II When thousand place ca be occupied by any of the digits out of 3, 4, 5 and 6 ; thousand place can be filled in 4 ways ; Hundred place can be filled in 6 ways; unit place can be filled in 4 ways. The required. no., using find principle of Multiplication =4×6×5×4=480 ∴Required no. = 80+480 =560.
3.	2	Total no. of letters = 10; No. of vowels = 4 (I occurs twice); Treat the 4 vowels together as one letters, we have 6+1=7 letters; The 7 letters can be arranged in 7! Ways since I occurs twice, the 4 vowels can be arranged in $\frac{4!}{2!}$ ays ∴ Required numbers = $7!\times\frac{4!}{2!}$ =60480
4.	2	Total no. of letters 8; number of vowels = 3 r occurs twice Total no. of arrangements when there is no restriction $=\frac{8!}{2!}$; when there vowels are together, regarding than as one letter, we have only $5=1=6$ letters ; These six letters can be arranged in $\frac{6!}{2!}$ ways (since *r* occurs twice) But the three vowels can be arranged amongst themselves in 3! ways Hence no. of arrangements when vowels are together $\frac{6!}{2!}3!$

		$\therefore$ Required number $=\frac{8!}{2!}-\frac{6!}{2!}3!=18000$
5.	2	No of Digits = 4, they can be arranged in 4! Ways out of 4! = 24 no. there are 3! no. where '0' occurs at thousand place i.e. they are 3 digit numbers. ; $\therefore$ Required no. of numbers = 4!-3!=18 Here 0 occurs of unit place in = 6 times ; 1 occurs of unit place in = 4 times 2 occurs at the units place = 4 times; 3 occurs at units place = 4 times. Sum of the digits at unit place $=6\times0+4\times1+4\times2+4\times3=24$; Sum of the digits at tens and Hundred's place = 24. ; Now 1 occurs at thousand place = 6 ways; 2 occurs at thousand place = 6 ways; 3 occurs at thousand place = 6 ways; Sum of the digits at thousand place $=6\times1+6\times2+6\times3=36$ Required Sum $=36\times10^3+24\times10^1+24\times10^0=38664$.
6.	2	The total number of possible outcomes is 6^4. The number of possible outcomes in which 2 does not appear on any dice is 5^4. $\therefore$the no. of possible outcomes in which at least one dice shows a 2 is $6^4-5^4=1296-625=671$
7.	2	Since each place of 10 digit number can be filled in 2 ways. $\therefore$ Total number of 10 digits numbers formed = 2^{10}.
8.	1	There are 5 instances when the pair of dice would yield a sum of 8. The instances are 6 + 2, 5 + 3, 4 + 4, 3 + 5, 2 + 6. Where 6, 5, 4, 3, 2 are the results of the first dice and 2, 3, 4, 5, 6 are the results of the second dice. The number of different sums yielded when a pair of dice are rolled is 36. Therefore the probability of sum of 8 occurring is $\frac{5}{36}$. Since we have to get 8 twice, the required prob. is $\frac{5}{36}\times\frac{5}{36}=\frac{25}{1296}$.
9.	4	The total number of different sums generated by a throw of three dice is $6^3=216$. The instances when we get a sum of 17 or 18 are $6+6+6, 6+6+5, 6+5+6, 5+6+6$. Therefore the number of favorable instance is 4. The required probability is $\frac{4}{216}=\frac{1}{54}$.
10.	1	If the $n=no.$ of sides of polygon of n–vertices. $\Rightarrow$ No. of diagonals $={}^nC_2-n=\frac{n(n-1)}{2}-n, \Rightarrow 44=\frac{n(n-1)}{2}-$ n, $\Rightarrow 44=\frac{(n-3)n}{2}, \Rightarrow n(n-3)=88, \Rightarrow n=11$.